EN29F040A@TSOP32.BIN analysis report

Executive identification

• Chip/dump size: Eon EN29F040A, 4-Mbit / 512 KiB x8 flash; file size 524,288 bytes / 0x80000.

• Likely role: IBM PC110 modem/fax board flash, RIOS Ver 1.04, with Panasonic MN195001 references.

• Confidence: High for board role; low/medium for CPU architecture.

• SHA-256: `add4cb4a3d17f2216ea97e7078f0ce820d25e07a4fadf2103d565f8c95f6ecb4`

• MD5: `a9f38ef86fba9d31285308fd71a6072b`

• Byte entropy: 4.3080 bits/byte.

Evidence from this dump

• Offset 0x00000 is the ASCII banner `RIOS Ver 1.04` followed by erased 0xFF space.

• Sparse but useful embedded strings include `RIOS SYSTEMS Co.,Ltd.`, `PANASONIC MN195001`, many `Ver ...` strings, and `PROGRAM LOADER Ver.1.00` near the top end.

• No PC 0x55AA option-ROM signature, FAT boot sector, DOS executable marker, or obvious filesystem signature was found.

• The file name and public PC110 dump notes identify it as the modem-board flash. The internal MN195001 string also fits a modem/fax function.

• The CPU architecture of this external flash image is not confirmed. The package includes a tentative 8051-like probe only as a way to inspect selected active regions, not as a claim of architecture.

Structural notes

• Active 4 KiB ranges (heuristic, excludes mostly 0x00/0xFF blocks): 0x20000-0x27FFF, 0x30000-0x66FFF, 0x6E000-0x6EFFF, 0x70000-0x71FFF

• Longest 0xFF runs: 0x14 len 131052, 0x71547 len 59801, 0x27264 len 36252, 0x66B64 len 30940, 0x6F000 len 3824

• Longest 0x00 runs: 0x46E5F len 247, 0x4750B len 201, 0x46077 len 63, 0x46217 len 63, 0x463B7 len 63

• PC option-ROM signatures found:

(none)

Interesting strings

• 0x000000: `RIOS Ver 1.04`

• 0x04A68A: `RIOS SYSTEMS Co.,Ltd.`

• 0x04A6A0: `PANASONIC MN195001`

• 0x057136: `Ver F55`

• 0x057C78: `Ver V08`

• 0x058E97: `lVer R17`

• 0x05B763: `Ver V09`

• 0x05B881: `Ver T28`

• 0x05CE17: `Ver Z12`

• 0x05D8AB: `Ver Z08`

• 0x05F52F: `Ver V14`

• 0x05FCE0: `Ver V20`

- 0x060D4D: `Ver V06`
- 0x061089: `Ver V19`
- 0x061A9D: `Ver D28`
- 0x06211A: `Ver V05`
- 0x062615: `Ver V04`
- 0x062DDC: `Ver R29|`
- 0x063FFB: `Ver 931n`
- 0x064EEF: `Ver RRR`
- 0x0665D5: `Ver JJ2`
- 0x06FEF0: `Ver. 1.04`
- 0x070029: `RIOS`
- 0x070035: ` Ver`
- 0x07FEE2: `PROGRAM LOADER Ver.1.00`

Included supplemental files

- `strings/EN29F040A_modem_board.strings.txt` - all printable strings found by a simple ASCII scanner.
- `hexdumps/EN29F040A_modem_board_selected_hexdumps.txt` - selected annotated hexdumps.
- `disassembly/EN29F040A_modem_board_tentative_8051_probe_disasm.txt`

Caveats

- This is static triage, not a full reverse-engineering project. Hardware register names, MCU peripheral vector names, and exact memory maps require board schematics, data books, or emulator instrumentation.
- The EN29 firmware especially should not be assumed to be 8051 code just because a probe file exists. The visible evidence supports modem/fax-board firmware/data, but not a confirmed instruction set.